USB to Rule Them All: One Manageability Interface for AI Platforms

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Panel Discussion

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Panel Purpose – Align on one Mgmt. Interface for AI Clusters

Problem Statement

- Manageability is a scaling bottleneck for AI racks
- Legacy sideband interfaces (I2C/SMBus, I3C, VDMs, RBT, SPI, GPIO, SGPIO, LTPI) are inefficient:
  - Low Bandwidth
  - High Pin Count
  - Routing and Signal Integrity issues
- Vendors solve discovery, update, recovery, debug differently - cost multiplies every generation
- Panel to identify actions needed to converge on one & open manageability interface for AI Clusters

Existing Manageability Protocol Specs

- MCTP – Management Component Transport Protocol
- PLDM / SPDM – device management & attestation
- NC-SI – network controller sideband
- Redfish – fleet-level management API

Converging on USB in OCP

- MCTP over USB
  - Generic USB transport for manageability
  - PLDM / SPDM for management, update & attestation
- OCP Recovery over USB
- Caliptra with USB support – USB native common silicon Root of Trust
- **OCP OBMF-ICP v1.0 over USB** – bridging the gap for all legacy manageability interfaces on Devices and Host CPU (GPIO, SPI, I2C, I3C, x86 IO, MMIO, UART, IPMI, …)